Take-Home Part I

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The Model and Data

Let $\{Y_i : i = 1, \ldots, n\}$ denote a set of response variables and let $\{x_i : i = 1, \ldots, n\}$ represent a corresponding set of non-decreasing covariate values. The model for the response variables is

$$Y_i = \mu_i + w_i \tag{1}$$

where

$$\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}. \tag{2}$$

In (2), $\alpha \geq 0$, $\beta \geq 0$, and $\gamma \geq 0$ are unknown parameters, and for $-1 < \theta < 1$,

$$w_i = \theta w_{i-1} + \epsilon_i, \tag{3}$$

where the ϵ_i are independent and identically distributed with $\epsilon_i \sim N(0, \sigma^2)$.

Q1

Consider a simplified model in which the w_i in (1) are iid $N(0, \sigma^2)$, i.e., a traditional additive error nonlinear regression model with constant variance. Fit this model to the project data using generalized least squares to estimate the parameters of the expectation function.

(a) and (b): Report GLS estimates of α, β, γ and the moment-based estimate of σ^2. Approximate 95% confidence intervals for α, β, γ.

Table 1: GLS estimates with approximate 95% confidence intervals.

		95% CI	
Parameter	Estimate	Lower	Upper
α	0.506	-0.331	1.343
β	1.222	0.211	2.233
γ	0.161	0.114	0.209

As noted in the solution to Question 1: The lower endpoint of the interval for $\hat{\alpha}$ is negative, which falls outside the scientifically meaningful ("desired") parameter space $\alpha \geq 0$, though it does not prevent μ_i from being defined. Also, the moment-based estimate of σ^2 is $\hat{\sigma}^2 = 1.015$.

(c) and (d): A plot with the estimated expectation function overlaid on the scatterplot along with an approximate 95% pointwise confidence band. A plot of standardized residuals against the covariate.

(c): As noted in the provided solution to Question 1: An approximate pointwise confidence band is constructed via the delta method, whose derivatives of μ_i with respect to (α, β, γ) are

$$\frac{\partial \mu_i}{\partial \alpha} = x_i^{\beta x_i^{-\gamma}}, \qquad \frac{\partial \mu_i}{\partial \beta} = \mu_i x_i^{-\gamma} \log x_i, \qquad \frac{\partial \mu_i}{\partial \gamma} = -\mu_i \beta (\log x_i)^2 x_i^{-\gamma}.$$

(d): Standardized residuals were computed using the hat-matrix correction $r_i/[\hat{\sigma}^2(1 - h_{i,i})]^{1/2}$, expression (9.40) of the Stat 520 notes, and are returned directly by `nonlin` via `fit1$stdres`. Note: the solution sheet cites (9.37), the general plug-in form; (9.40) is its constant-variance simplification, obtained by substituting $\hat{\sigma}_i^2 = \hat{\sigma}^2$. The residual plot shows a good deal of remaining structure as a function of the covariate, consistent with Figure 3 of Sibbesen (1981). This pattern motivate the use of an autoregressive structure in the "regression equation error terms", i.e., the inclusion of an AR(1) small-scale component in the full model, treating the serial dependence across ordered covariate values analogously to serial correlation over time.

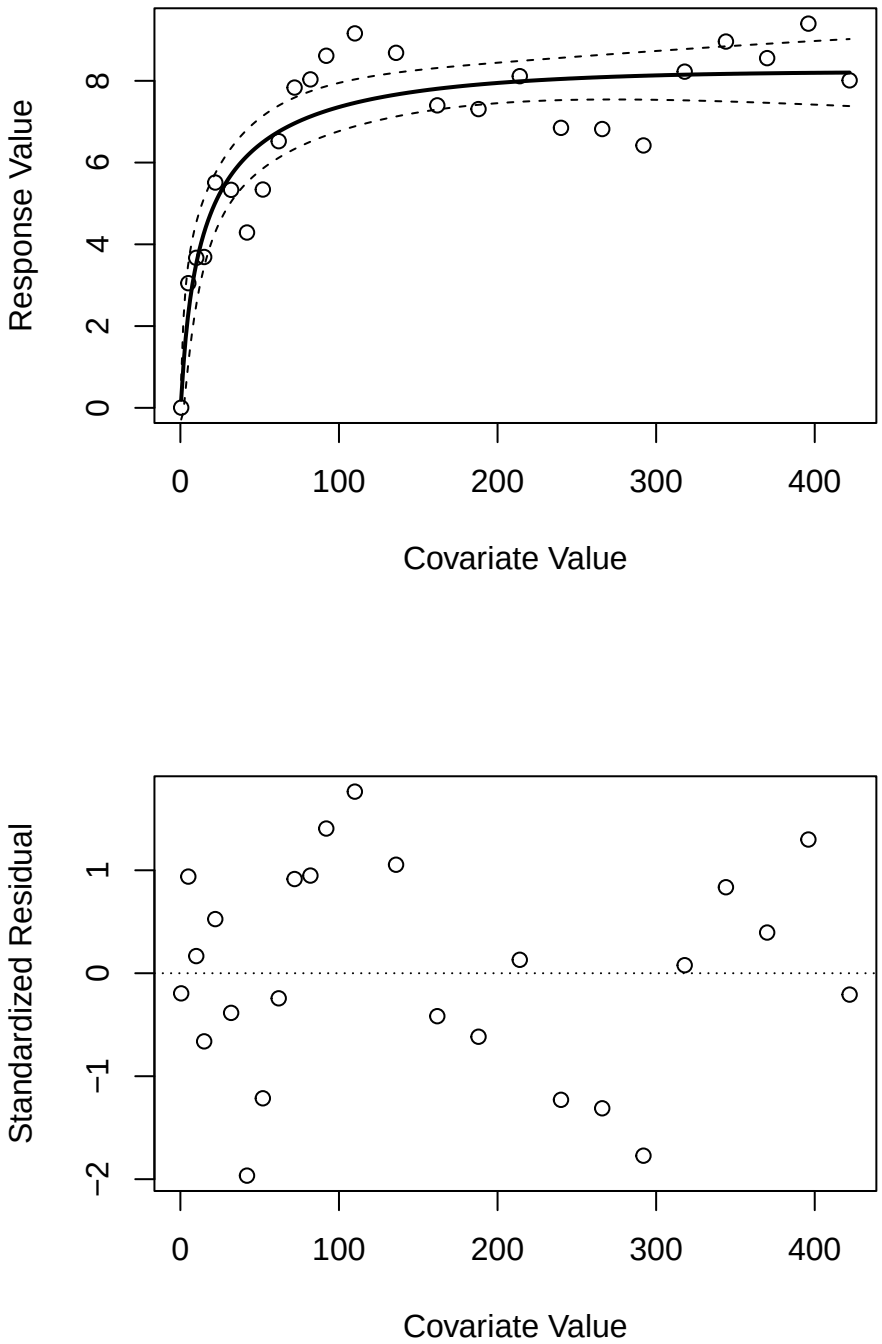

Figure 1: (Top) Fitted extended Freundlich curve with approximate 95% pointwise confidence band. (Bottom) Standardized residuals versus covariate.

Q2

Consider now just the portion of the model for w_i in (3). Suppose we were able to observe the w_i. Take $W_0 = 0$, and use without derivation that under second-order stationarity, $E(W_i) = 0$ and $E(W_i^2) = \sigma^2/(1-\theta^2)$ for all i.

(a) Show that the estimating equation for θ based on the naive estimating function

$$v_i(\mathbf{W}, \theta) = W_i - \theta W_{i-1}$$

is an unbiased estimating equation.

The naive estimating function is $v_i(\mathbf{W}, \theta) = W_i - \theta W_{i-1}$. Substituting $W_i = \theta W_{i-1} + \epsilon_i$ directly gives

$$E[v_i(W, \theta)] = E[\theta W_{i-1} + \epsilon_i - \theta W_{i-1}] = E[\epsilon_i] = 0,$$

since $\epsilon_i \sim N(0, \sigma^2)$ is independent of W_{i-1}. Because this holds for every i and all $-1 < \theta < 1$, the estimating equation $\sum_{i=2}^n v_i(\mathbf{W}, \theta) = 0$ is unbiased. Setting it to zero and solving gives

$$\tilde{\theta}_{\text{naive}} = \frac{\sum_{i=2}^n W_i}{\sum_{i=2}^n W_{i-1}}.$$

While this form is algebraically valid, the denominator $\sum W_{i-1}$ can be near zero, possibly contributing to poor numerical behavior seen in Q2(d), **spoilers**.

(b) Show that the estimating equation for θ based on the optimal estimating function

$$v_i(\mathbf{W}, \theta) = W_{i-1}(W_i - \theta W_{i-1})$$

is an unbiased estimating equation.

The optimal estimating function is $v_i(\mathbf{W}, \theta) = W_{i-1}(W_i - \theta W_{i-1})$. Expanding the expectation and using the model $W_i = \theta W_{i-1} + \epsilon_i$,

$$E[W_{i-1}W_i] = E[W_{i-1}(\theta W_{i-1} + \epsilon_i)] = \theta E[W_{i-1}^2] + E[W_{i-1}\,\epsilon_i].$$

Because W_{i-1} is a function of $\epsilon_1, \ldots, \epsilon_{i-1}$ alone and ϵ_i is independent of those, $E[W_{i-1}\epsilon_i] = 0$.

Under the assumption of second-order (weak) stationarity of $\{W_i\}$ — properties as given in the initial problem statement, in addition to and $\text{Cov}(W_i, W_j)$ depending only on $|i-j|$, which holds when $|\theta| < 1$ — we have $E[W_{i-1}^2] = \sigma^2/(1-\theta^2)$,

$$E[v_i(\mathbf{W}, \theta)] = \theta\frac{\sigma^2}{1-\theta^2} - \theta\frac{\sigma^2}{1-\theta^2} = 0.$$

The estimating equation $\sum_{i=2}^n W_{i-1}(W_i - \theta W_{i-1}) = 0$ is therefore unbiased, and its solution is the lag-1 autocorrelation estimator

$$\tilde{\theta}_{\text{opt}} = \frac{\sum_{i=2}^n W_i W_{i-1}}{\sum_{i=2}^n W_{i-1}^2}.$$

This is called "optimal" in the sense of Godambe's criterion: Among linear estimating functions in W_{i-1}, multiplying the naive function by W_{i-1} maximizes the ratio $[E\{\partial v/\partial\theta\}]^2/\operatorname{var}\{v\}$ (see Ch. 8.3, the "measure of the lost efficiency of the solution to any estimating function"). Note that both estimating equations yield closed-form estimators, and the two estimators are in general numerically distinct.

(c) Develop a moment-based estimator for σ^2 that could be used with either estimating equation, and show that it is unbiased.

Because $\epsilon_i = W_i - \theta W_{i-1}$ and $\epsilon_i \sim N(0, \sigma^2)$, we have $E[\epsilon_i^2] = \sigma^2$ for all i. Plugging in $\tilde{\theta}$ from either estimating equation and using $W_0 = 0$ would motivate a possible estimator of the form

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(W_i - \tilde{\theta}\, W_{i-1})^2,$$

where $W_0 = 0$. Note: The $i = 1$ term contributes $W_1 - \tilde{\theta}W_0 = W_1$, which equals ϵ_1 by virtue of the boundary condition $W_0 = 0$; this is a boundary artifact rather than a general property of the innovation structure. To see that this estimator is approximately unbiased, note that at the true θ each term satisfies $E[(W_i - \theta W_{i-1})^2] = E[\epsilon_i^2] = \sigma^2$, so

$$E\left[\frac{1}{n}\sum_{i=1}^{n}(W_i - \theta W_{i-1})^2\right] = \sigma^2.$$

Once θ is replaced by $\tilde{\theta}$, however, strict unbiasedness no longer holds: $E[(W_i - \tilde{\theta}W_{i-1})^2] \neq \sigma^2$ exactly, because $\tilde{\theta}$ is itself a function of the data. The estimator is therefore only approximately unbiased, in the same spirit as s^2 in ordinary regression, with the $n-1$ denominator providing a degrees-of-freedom correction by analogy with the linear case — though unlike s^2 in ordinary regression, this correction does not achieve exact unbiasedness here because the innovation residuals are not linear projections in the standard sense.

(d) Apply both estimating equations using the raw residuals from Q1 as surrogate observations of w_i. Compute $\hat{\sigma}^2$ for each and compare all three estimates of σ^2 (naive EE, optimal EE, Q1 GLS). Are these estimators all estimating the same quantity?

Table 2: Comparison of $\hat{\theta}$ and $\hat{\sigma}^2$ estimates. Note that $\tilde{\theta}_{\text{naive}} > 1$ falls outside the desired parameter space $|\theta| < 1$.

Method	$\hat{\theta}$	$\hat{\sigma}^2$
Naive EE	1.0046	0.8559
Optimal EE	0.5422	0.6573
Q1 GLS	—	1.0149

Because the Q1 residuals $\hat{\epsilon}_i$ contain mean-estimation error, i.e., they are $y_i - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ rather than $w_i = y_i - \mu_i(\alpha, \beta, \gamma)$, the estimating equations applied here are only approximations to those derived in Q2(a) and (b). The two estimating equations yield different estimates of θ, though both target the same population parameter, that parameter being the AR(1) autocorrelation coefficient θ in model (3). The numerical difference between $\tilde{\theta}_{\text{naive}} = 1.005$ and $\tilde{\theta}_{\text{opt}} = 0.542$ reflects estimation error rather than any difference in what the equations are targetting, with the optimal estimate being the more reliable of the two (lower asymptotic variance by Godambe's criterion), perhaps reflecting its naming convention.

More importantly, the *three* $\hat{\sigma}^2$ values are **not** all estimating the same quantity. The Q1 GLS estimate was obtained under the assumption that the w_i are iid, so it targets $\mathrm{Var}(w_i)$; because the true model is AR(1), or at least we are motivated to believe it is AR(1), $\mathrm{Var}(w_i) = \sigma^2/(1-\theta^2)$. However, the Q1 GLS estimator itself has no knowledge of that structure. In contrast, the two EE-based estimates are obtained by squaring residuals of the form $W_i - \tilde{\theta} W_{i-1} \approx \epsilon_i$, so they target the innovation variance σ^2. Because the AR(1) process has positive autocorrelation ($\theta > 0$ appears typical for sorption data), the marginal variance exceeds the innovation variance, and we expect $\hat{\sigma}^2_{\mathrm{Q1}} > \hat{\sigma}^2_{\mathrm{naive}} \approx \hat{\sigma}^2_{\mathrm{opt}}$.

Q3

Assume $x_0 = 0$ and $w_0 = 0$ (based on Q1, Q2). Derive the full log likelihood for the model of (1), (2), and (3), and maximize it to find simultaneous estimates of α, β, γ, θ, and σ^2.

(a) An expression for the log likelihood.

Given $w_0 = 0$, the residuals $w_i = Y_i - \mu_i$ satisfy $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. The likelihood derived here is a *conditional* likelihood, conditioning on $w_0 = 0$ rather than treating W_1 as drawn from the stationary marginal distribution $N(0, \sigma^2/(1-\theta^2))$. Given this conditioning, each innovation $\epsilon_i = w_i - \theta w_{i-1}$ is independent, so the joint density of $(Y_1, \ldots, Y_n)$ may be written as a product of n normal densities.

Under these conditions, the log likelihood may be written

$$\ell(\alpha, \beta, \gamma, \theta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(w_i - \theta w_{i-1}\big)^2,$$

where $w_i = Y_i - \mu_i = Y_i - \alpha x_i^{\beta x_i^{-\gamma}}$ and $w_0 = 0$.

(b) A description of how you maximized the log likelihood. Note that Q1 estimates provide natural starting values and $\theta_0 = 0$; `optim` with `method = "BFGS"` is recommended over Nelder-Mead for five parameters.

Wald

The log likelihood was maximized using `optim` with `method = "BFGS"`. Starting values for (α, β, γ) were taken from the Q1 GLS fit; $\theta_0 = 0$ and $\sigma_0^2 = \hat{\sigma}^2_{\mathrm{Q1}}$ completed the five-parameter starting vector. These are natural starting values given that the Q1 fit is the $\theta = 0$ special case of the full model, placing the initial iterate close to the region of the likelihood surface where the solution is expected to lie and reducing sensitivity to the starting value concern noted for five-parameter problems of this type. Because `optim` minimizes by default, we pass the negative log likelihood. The Hessian was then extracted by specifying `hessian = TRUE` for Wald inference in part (c).

Profile Method

Profile likelihood confidence intervals were constructed to complement the Wald intervals, as Wald intervals can perform poorly when the log-likelihood is asymmetric or the MLEs are near parameter boundaries—both plausible concerns inthis application. For each parameter $\psi \in \{\alpha, \beta, \gamma, \theta, \sigma^2\}$, the profile log-likelihood is

$$\ell_p(\psi) = \max_{\boldsymbol{\lambda}} \ \ell(\psi, \boldsymbol{\lambda}),$$

where $\boldsymbol{\lambda}$ denotes the remaining four parameters. At each point of a 60-point grid spanning $\hat{\psi} \pm 4$ Wald standard errors—where the Wald SE is taken from the observed information matrix at the MLE—truncated to the parameter's natural support, the inner maximization over $\boldsymbol{\lambda}$ was performed using `optim` with `method = "BFGS"`, initialized at the full MLE. Nelder-Mead was also evaluated as a robustness check against cases where constraint boundaries cause finite-difference gradient evaluations to return non-finite values; both methods converged to equivalent results, and BFGS was retained. The 95% confidence interval for ψ is the set of grid values satisfying

$$\ell_p(\psi) \geq \hat{\ell} - \tfrac{1}{2}\chi^2_{0.95,\,1},$$

with endpoints obtained by linear interpolation at the threshold crossing. The choice to profile over single parameters is one of convenience; profiling over pairs or higher-dimensional subsets is equally valid but produced negligible differences in preliminary analysis.

(c) The MLEs, 95% confidence intervals (stating whether Wald or profile), and the maximized log likelihood value.

Wald

Wald confidence intervals were constructed from the observed information matrix — the negative Hessian of ℓ evaluated at the MLE, returned directly by `optim` via `hessian = TRUE` — used in place of the expected Fisher information $\mathcal{I}(\boldsymbol{\beta}) = -E(\nabla^2\ell)$. These are asymptotically equivalent under standard regularity conditions, though the distinction is worth noting: with $n = 23$ and $p = 5$, these intervals rely heavily on asymptotic normality and should be interpreted with caution.

Profile Method

Profile 95% confidence intervals are reported alongside the Wald intervals in Table 3. The most consequential difference is for α: the Wald lower bound of -0.4834 is negative and therefore inadmissible under the model, which requires $\alpha \geq 0$. The profile lower bound of 0.0801 correctly respects this constraint, reflecting the asymmetry of the likelihood surface near the boundary. Right-skew is also evident for σ^2. Where the two interval types agree closely — as for γ and θ — the likelihood surface is approximately quadratic and the Wald approximation is adequate. Given the inadmissibility of the Wald interval for α, profile confidence intervals are used as the basis for all subsequent inference.

Profile log-likelihood maxima matched $\hat{\ell} = -29.598$ to within 0.001 across all parameters, confirming that the inner BFGS optimization successfully recovered the MLE when each parameter was held fixed at its estimated value. The one negative difference for θ (-0.0001) is numerical noise from the inner optimizer slightly overshooting, not a genuine improvement over the MLE.

The profile upper bound for θ of 0.9151 warrants note: it lies close to the stationarity boundary $|\theta| = 1$, meaning the data are consistent with near-unit-root serial dependence. Under near-unit-root conditions, the marginal variance $\sigma^2/(1-\theta^2)$ becomes large and poorly estimated, and the conditional likelihood approximation based on $w_0 = 0$ may be less accurate than asymptotic theory suggests. This is a limitation that cannot be resolved at $n = 23$.

Table 3: Wald and profile 95% confidence intervals at the MLE. Maximized log likelihood: -29.5981.

		Wald 95% CI		Profile 95% CI	
Parameter	MLE	Lower	Upper	Lower	Upper
α	0.7395	-0.4834	1.9625	0.0801	2.2498
β	1.0283	0.0275	2.0291	0.3854	2.0480
γ	0.1574	0.0929	0.2219	0.0682	0.2188
θ	0.5598	0.2203	0.8993	0.2102	0.9152
σ^2	0.6250	0.2785	0.9714	0.3763	1.1510

(d) A plot of the data with the estimated expectation function overlaid along with an approximate 95% pointwise confidence band.

The confidence band for $\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}$ uses the delta method with the 3×3 submatrix of $\widehat{\text{Cov}}$ corresponding to (α, β, γ). The derivatives are the same as in Q1.

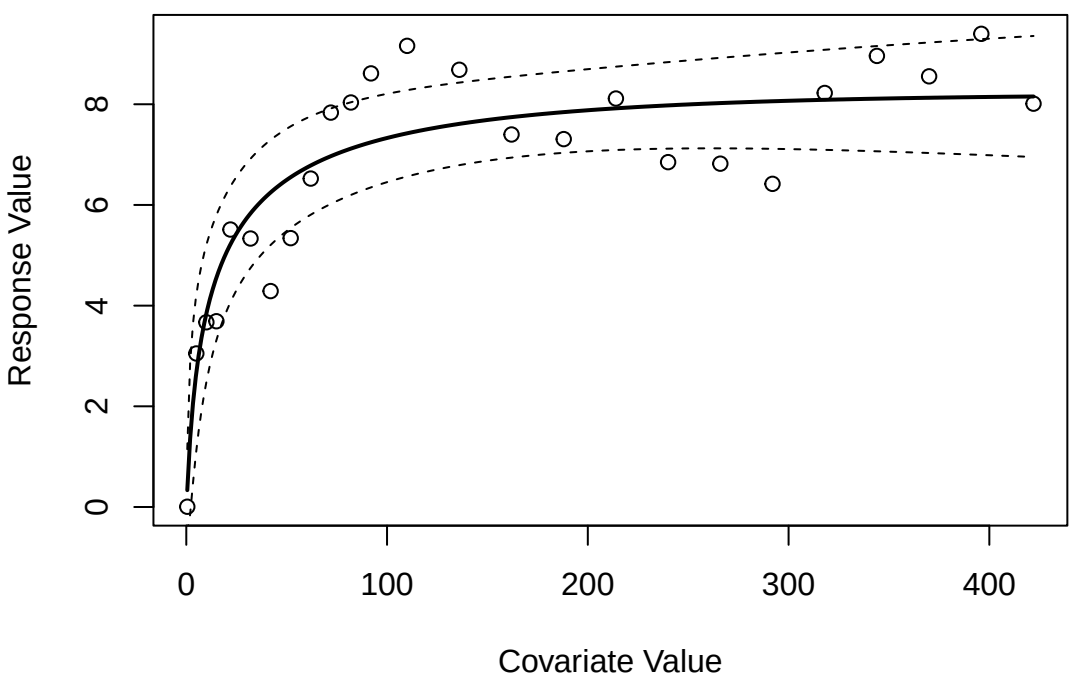

Figure 2: Fitted extended Freundlich curve with approximate 95% pointwise confidence band via the delta method. Scalar parameter CIs in (c) use profile intervals; the band uses the Wald delta method as profile bands for functions of multiple parameters are not standard.

(e) A comparison of the large-scale component (α, β, γ) to Q1: point estimates, interval estimates, estimated correlations among parameter estimators, and confidence band widths. Discuss any differences.

Table 4: Point estimates and 95% CI widths by method (Q1 Wald, Q3 profile). Mean confidence band half-width: Q1 = 0.6225, Q3 = 0.9666.

	Point Estimate		CI Width	
Parameter	Q1 GLS	Q3 MLE	Q1 GLS	Q3 Profile
α	0.5063	0.7395	1.6737	2.1697
β	1.2220	1.0283	2.0214	1.6625
γ	0.1614	0.1574	0.0944	0.1506

Table 5: Estimated correlation matrix for (α, β, γ): Q1 GLS.

	α	β	γ
α	1.000	-0.987	-0.837
β	-0.987	1.000	0.914
γ	-0.837	0.914	1.000

Table 6: Estimated correlation matrix for (α, β, γ): Q3 MLE.

	α	β	γ
α	1.000	-0.968	-0.685
β	-0.968	1.000	0.843
γ	-0.685	0.843	1.000

The point estimates of (α, β, γ) shift noticeably between Q1 and Q3 — most visibly $\hat{\alpha}$ moving from 0.5063 to 0.7395 — because jointly estimating θ reweights the observations according to the AR(1) correlation structure rather than treating them as independent. When correlation parameters are known, GLS with $\Sigma^{-1}(\theta)$ as a fixed weight matrix improves efficiency over the iid estimator. When they must be estimated jointly, however, the additional uncertainty in θ and σ^2 propagates into the regression parameters and can widen confidence intervals.

The profile CI widths under Q3 are wider than Q1 Wald for α and γ, narrower for β, and the mean confidence band half-width increases from 0.6225 to 0.9666. The inadmissible Wald lower bound of -0.4834 for α is replaced by a profile lower bound of 0.0801, correctly respecting the $\alpha > 0$ constraint. This reflects the additional uncertainty introduced by estimating θ and σ^2 jointly with the regression parameters: the likelihood surface is more diffuse in the (α, β, γ) directions once the correlated error structure is acknowledged, and the Q1 covariance matrix, which ignored this coupling, was understating that uncertainty. It should also be noted that Table 4 compares Q1 Wald intervals to Q3 profile intervals — methodologically distinct constructions. Profile intervals are generally wider than Wald when the likelihood surface is asymmetric, so part of the observed width difference reflects the interval type rather than purely the model specification change.

This effect is partially offset by the fact that $\hat{\sigma}^2_{\text{Q1}} = 1.015 > \hat{\sigma}^2_{\text{MLE}} = 0.625$, which inflates the Q1 standard errors; the Q3 intervals would be even wider if σ^2 were held equal across methods. The estimated correlations

among the regression parameter estimators also shift under Q3, with $|\hat{\rho}_{\alpha,\beta}| \approx 0.97$ under both methods indicating that the likelihood surface is extremely flat along the (α, β) ridge — meaning individual parameter estimates are highly sensitive to small perturbations regardless of the fitting method, and the wide confidence intervals are intrinsic to the model structure rather than an artifact of the estimation procedure.

(f) A comparison of the small-scale component (θ, σ^2) to the optimal estimating equation results from Q2. Two estimates of θ and three estimates of σ^2 are available (Q3 MLE, Q1 GLS, Q2 moment-based). Discuss any differences.

Table 7: Comparison of small-scale parameter estimates.

Quantity	Q2 Optimal EE	Q3 MLE
θ	0.5422	0.5598
σ^2	0.6573	0.6250

Table 8: All estimates of σ^2 across methods.

Estimator	$\hat{\sigma}^2$
Q1 GLS (marginal variance)	1.0149
Q2 moment-based	0.6573
Q3 MLE	0.6250

The two estimates of θ agree closely: the Q2 optimal estimating equation yields $\hat{\theta} = 0.5422$ and the Q3 MLE yields $\hat{\theta} = 0.5598$. Both target the same parameter but differ in how they handle the regression function. The Q2 estimate used raw residuals from the misspecified iid model of Q1 as surrogates for w_i, introducing noise from the first-stage estimation; the Q3 MLE estimates θ simultaneously with (α, β, γ), which should reduce that bias. The small discrepancy is consistent with this explanation.

For σ^2, the three estimators target different quantities, as established in Q2(d): the Q1 GLS estimate of 1.0149 targets the marginal variance $\sigma^2/(1-\theta^2)$ and is therefore expected to be the largest when $\theta > 0$. Both the Q2 moment-based estimate (0.6573) and the Q3 MLE (0.6250) target the innovation variance σ^2, and are indeed substantially smaller. The Q2 version inherits approximation error from using Q1 residuals instead of true w_i values, while the Q3 MLE is obtained from the correctly specified joint model; the remaining discrepancy of 0.0323 between them is attributable to that first-stage approximation.

Q4

Assume $x_0 = 0$ and $w_0 = 0$. Using the events $A_i = (Y_i = y_i \mid Y_{i-1} = y_{i-1})$ for $i = 2, \ldots, n$, derive a pairwise conditional log composite likelihood and fit the model. Report the same components as in Q3.

Additionally, discuss which asymptotic context for composite likelihood is most appropriate given $n = 23$. Address potential issues with estimating the H and J matrices in the Godambe information; compute them if possible and use them to construct interval estimates for the regression function parameters.

Pairwise Conditional Log Composite Likelihood

Given the model, $w_i = Y_i - \mu_i$, and the AR(1) structure $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \sim N(0, \sigma^2)$, the conditional distribution of Y_i given $Y_{i-1} = y_{i-1}$ is

$$Y_i \mid Y_{i-1} = y_{i-1} \sim N\big(\mu_i + \theta(y_{i-1} - \mu_{i-1}),\ \sigma^2\big),$$

since $w_i \mid w_{i-1} \sim N(\theta w_{i-1}, \sigma^2)$. Each conditional log likelihood piece is therefore

$$\ell_i = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\sigma^2 - \frac{\epsilon_i^2}{2\sigma^2}, \qquad \epsilon_i = (y_i - \mu_i) - \theta(y_{i-1} - \mu_{i-1}),$$

and the pairwise conditional log composite likelihood is

$$\ell_C = \sum_{i=2}^{n} \ell_i = -\frac{n-1}{2}\log(2\pi) - \frac{n-1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=2}^{n}\epsilon_i^2.$$

The composite likelihood thus conditions on Y_1 (i.e., sums from $i = 2$), whereas the full likelihood in Q3 used $w_0 = 0$ to include $i = 1$, so the two differ by exactly one term.

Asymptotic Context

This problem falls under Asymptotic Context II (Ch. 9.3.2): a single realization of a dependent sequence of length $n = 23$, not independent replicates of a multivariate vector. Under Context II, the Comets–Janzura result would require translation-invariant conditional specification, which fails here because μ_i varies with the covariate. The Context I estimators of H and J—averaging over iid replicates—are therefore unavailable. We instead compute empirical analogs: $\hat{H}$ from the negative Hessian of ℓ_C at the MCLE, and $\hat{J}$ from the sum of outer products of piece scores $\nabla\ell_i$. With $n = 23$ these should be interpreted cautiously; parametric bootstrap would be the principled alternative. Godambe-based intervals are reported as the analytic option, with this caveat noted.

(a) Expression for the Log Composite Likelihood

See derivation above; implemented in `logcl_q4`.

(b) Maximization

The log composite likelihood was maximized using `optim` with `method = "BFGS"`, using Q1 GLS estimates as starting values for (α, β, γ) and $\theta_0 = 0$, $\sigma_0^2 = \hat{\sigma}^2_{\text{Q1}}$.

(c) MCLEs, Confidence Intervals, and Maximized Log Composite

Likelihood Wald intervals use the standard inverse-Hessian covariance, computed from the Hessian of the negative composite log likelihood returned by `optim`. These are reported alongside the Godambe-based intervals further below.

Table 9: MCLEs with standard errors and Wald 95% confidence intervals. Maximized log composite likelihood: -28.7688.

			95% CI	
Parameter	MCLE	SE	Lower	Upper
α	1.0628	1.3946	-1.6705	3.7960
β	0.8126	0.7692	-0.6951	2.3202
γ	0.1448	0.0585	0.0302	0.2594
θ	0.5752	0.1832	0.2161	0.9343
σ^2	0.6437	0.1858	0.2795	1.0080

H and J Matrices and Godambe-Based Intervals

For $\hat{J}$ we compute the sum of outer products of individual piece score vectors $\nabla \ell_i$, approximated by numerical central differences. This uses the Context I outer-product form (Ch. 9.3.1); under the Comets-Janzura result for Context II, J_m would instead involve neighborhood-based sums (Ch. 9, eq. 9.34), but since that result does not apply here, the Context I formula is used as a practical approximation with no direct theoretical justification under Context II. The resulting Godambe intervals should therefore be interpreted as heuristic rather than formally valid. Examining Tables 11 and 12, the $(3,3)$ elements of both $\hat{H}$ (38593.78) and $\hat{J}$ (40035.62) are very large relative to all others, reflecting high curvature in the γ direction and suggesting γ is the most precisely determined parameter locally. The large negative off-diagonal entries in rows/columns 1 and 3, and 2 and 3, are consistent with the high (α, γ) and (β, γ) correlations seen in Table 15, and confirm that the identifiability challenges in this model are concentrated in the (α, β, γ) block.

Table 10: MCLEs with Godambe-based 95% confidence intervals.

			95\% CI	
Parameter	MCLE	SE	Lower	Upper
α	1.0628	0.9597	-0.8182	2.9437
β	0.8126	0.5672	-0.2991	1.9242
γ	0.1448	0.0498	0.0472	0.2425
θ	0.5752	0.1787	0.2250	0.9254
σ^2	0.6437	0.1213	0.4061	0.8814

Table 11: Estimated H matrix (negative Hessian of composite log likelihood at MCLE).

	α	β	γ	θ	σ^2
α	328.3120	831.6470	-3442.0950	-2.2842	0.0257
β	831.6470	2122.1757	-8903.9726	3.2073	-0.0161
γ	-3442.0950	-8903.9726	38593.7805	-61.7570	0.3338
θ	-2.2842	3.2073	-61.7570	36.4902	-0.0082
σ^2	0.0257	-0.0161	0.3338	-0.0082	28.9569

Table 12: Estimated J matrix (sum of outer score products at MCLE).

	α	β	γ	θ	σ^2
α	332.8818	851.2476	-3562.8526	-5.8426	-18.2499
β	851.2476	2184.9056	-9229.8893	-12.2185	-44.2914
γ	-3562.8526	-9229.8893	40035.6206	31.7402	153.9595
θ	-5.8426	-12.2185	31.7402	29.5949	1.7960
σ^2	-18.2499	-44.2914	153.9595	1.7960	12.3473

(d) Fitted Curve and 95% Pointwise Confidence Band

The band uses the delta method with the (α, β, γ) block of the Godambe covariance $\hat{G}^{-1}$.

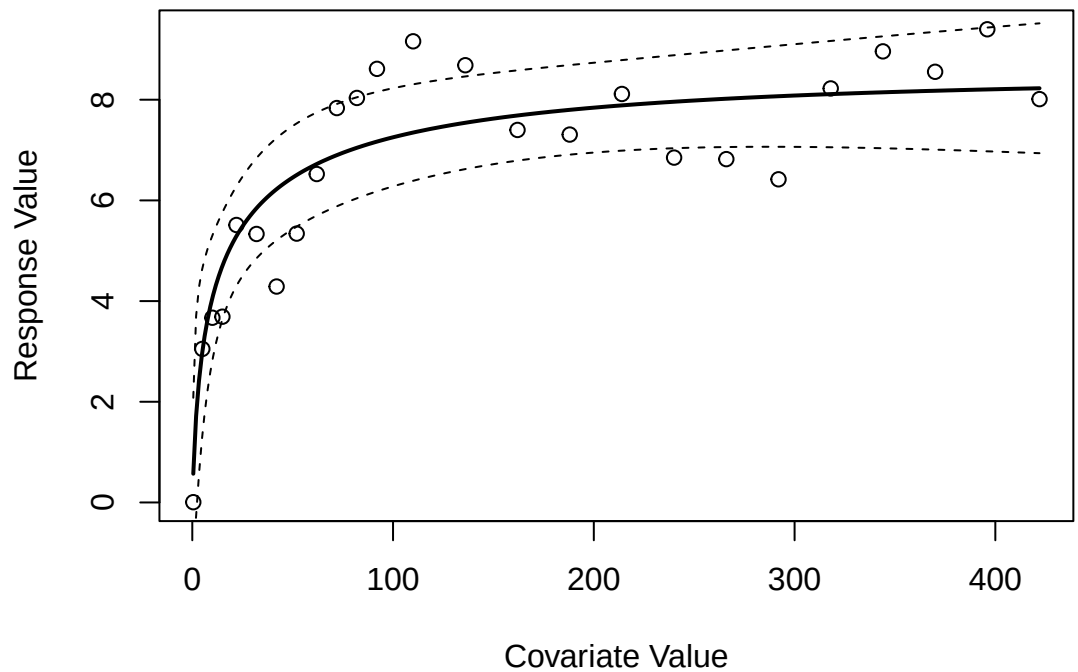

Figure: Fitted extended Freundlich curve with approximate 95% pointwise confidence band based on the Godambe covariance. Band uses the delta method with the (α, β, γ) block of $\hat{G}^{-1}$; see Q3(d) for the analogous Wald construction.

(e) Comparison of Large-Scale Component to Q1 and Q3

Table 13: Point estimates of large-scale parameters across methods.

Parameter	Q1 GLS	Q3 MLE	Q4 MCLE
α	0.5063	0.7395	1.0628
β	1.2220	1.0283	0.8126
γ	0.1614	0.1574	0.1448

Table 14: 95% CI widths for large-scale parameters across methods (Q1 Wald, Q3 profile, Q4 Wald and Godambe). Mean confidence band half-width (Q4): 1.0281.

			Q4	
Parameter	Q1 GLS	Q3 MLE	Q4 Wald	Q4 Godambe
α	1.6737	2.1697	5.4666	3.7620
β	2.0214	1.6625	3.0152	2.2234
γ	0.0944	0.1506	0.2292	0.1953

Table 15: Estimated correlation matrix for (α, β, γ): Q4 Godambe.

	α	β	γ
α	1.000	-0.980	-0.845
β	-0.980	1.000	0.932
γ	-0.845	0.932	1.000

The point estimates show a clear monotone progression across methods: $\hat{\alpha}$ increases from 0.506 (Q1) to 0.740 (Q3) to 1.063 (Q4), while $\hat{\beta}$ and $\hat{\gamma}$ decrease correspondingly. This pattern is consistent with movement along the likelihood ridge defined by the near-collinearity among the large-scale parameters — Table 15 shows $\hat{\rho}_{\alpha,\beta} = -0.980$ and $\hat{\rho}_{\alpha,\gamma} = -0.845$ under Q4, so small shifts in the likelihood surface translate into substantial opposing movements in individual parameter estimates. The progression is not noise; it reflects the successive reduction in the number of terms contributing to the likelihood (Q1 ignores θ entirely, Q3 conditions on $w_0 = 0$ using all n terms, Q4 conditions on Y_1 using only $n - 1$ terms).

The CI widths (Table 14) tell a nuanced story. Q4 Wald intervals are the widest of all four methods for every parameter — substantially so for α (5.467 under Q4 Wald vs 2.044 under Q3 profile) — because the inverse-Hessian covariance of the composite likelihood inherits the near-singular curvature of the five-parameter surface without the stabilizing contribution of the first observation. The Godambe-based intervals are narrower than Q4 Wald for all three parameters, because $\hat{J} \approx \hat{H}$ numerically (Tables 11 and 12 are in close agreement), so the sandwich correction $\hat{H}^{-1}\hat{J}\hat{H}^{-1}$ deflates slightly relative to $\hat{H}^{-1}$ alone rather than inflating as it would when score variability substantially exceeds curvature. It is worth noting, however, that this agreement may partly reflect the small number of outer products ($n - 1 = 22$) used to estimate $\hat{J}$, which may not be sufficient to reliably distinguish J from H at this sample size. Under correct specification of the composite likelihood, $J = H$ is expected to hold asymptotically—so the approximate equality $\hat{J} \approx \hat{H}$ provides informal evidence that the pairwise conditional model is not grossly misspecified, though the small sample makes this interpretation tentative.

Turning to the standardized residual plots, all three methods produce visually similar patterns — the residuals are confined roughly to $(-2, 1)$ across the covariate range, with no obvious trend or fan. This is

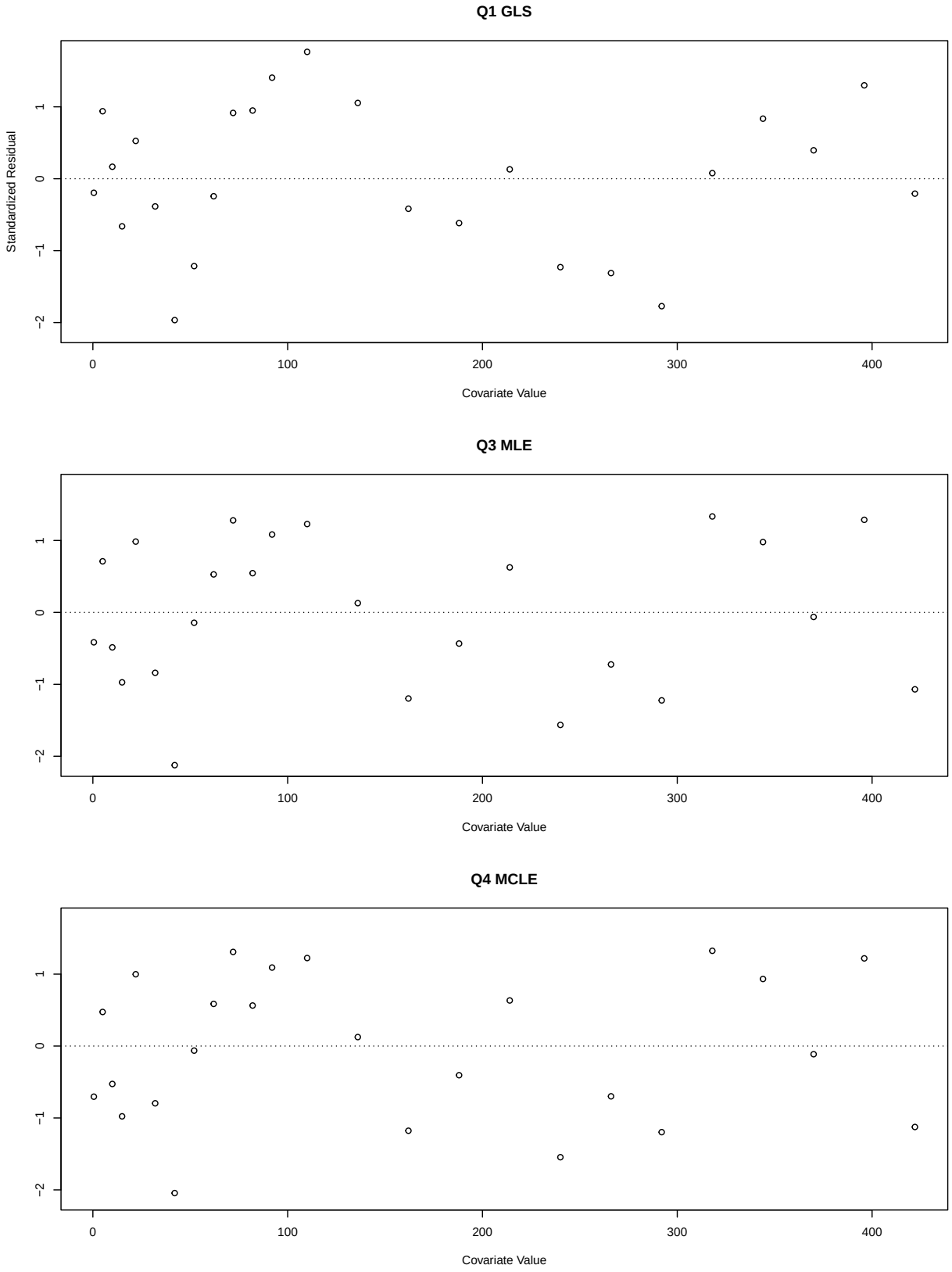

Figure 3: Standardized residuals versus covariate under Q1 GLS (left), Q3 MLE (center), and Q4 MCLE (right).

somewhat surprising given how different the Q1 and Q3/Q4 model specifications are, and warrants caution in interpretation. Two observations: first, the innovation residuals used for Q3 and Q4 ($\hat{\epsilon}_i = \hat{w}_i - \hat{\theta}\hat{w}_{i-1}$) are conceptually different objects from the Q1 standardized residuals — they have had the AR(1) dependence explicitly removed, so the absence of trend in the Q3/Q4 panels is the expected behavior of correctly whitened residuals, not evidence that Q1 was adequate. Second, there does appear to be mild cyclical structure across all three panels — positive residuals cluster at low and high covariate values, with a slight dip in the mid-range — which may reflect a small remaining misspecification in the mean function μ_i itself rather than the error structure. This would be consistent with Sibbesen's (1981) observation that the extended Freundlich curve is an empirical approximation and may not perfectly capture the sorption mechanism at all covariate values. With $n = 23$ it is not possible to formally test for this, but it is a limitation worth noting. Plotting residuals against $\sqrt{x}$ or $\log(x)$ in place of x compresses the dense cluster of small covariate values and can make structure in that region easier to see, but does not alter the residuals themselves or change any of the above comparisons — the same patterns are present under any monotone transformation of the horizontal axis, and the substantive conclusions about whitening, cyclical structure, and mean misspecification are unaffected by the choice of scale.

(f) Comparison of Small-Scale Component

Table 16: Comparison of small-scale parameter estimates across methods.

Quantity	Q2 Optimal EE	Q3 MLE	Q4 MCLE
θ	0.5422	0.5598	0.5752
σ^2	0.6573	0.6250	0.6437

Table 17: All estimates of σ^2 across methods.

Estimator	$\hat{\sigma}^2$
Q1 GLS (marginal variance)	1.0149
Q2 moment-based	0.6573
Q3 MLE	0.6250
Q4 MCLE	0.6437

The Q4 estimates of θ and σ^2 are close to Q3, as expected given that conditioning on Y_1 rather than fixing $w_0 = 0$ differs by a single term. The small monotone progression $\tilde{\theta}_{\text{opt}} = 0.542 \to \hat{\theta}_{\text{Q3}} = 0.560 \to \hat{\theta}_{\text{Q4}} = 0.575$ is consistent with successive reduction of first-stage approximation error: Q2 uses misspecified residuals, Q3 jointly estimates all parameters from the full conditional likelihood, and Q4 does the same from the pairwise conditional likelihood which down-weights the first observation. Any divergence again reflects the single omitted term. The same σ^2 hierarchy established in Q2(d) and Q3(f) applies: Q1 GLS targets $\sigma^2/(1-\theta^2)$, while Q2, Q3, and Q4 all target the innovation variance σ^2.

Q5

A two-stage estimation strategy would consist of estimating α, β, γ from the nonlinear constant variance model of Q1, and then using the residuals from that fit to estimate θ and σ^2 via the optimal estimating equation of Q2. Based on what you have learned in completing this exam, what would you conclude about the possibility of using such a two-stage procedure to fit the overall extended Freundlich model to data from this problem?

Overall

The two-stage procedure is superficially appealing because it decouples a hard five-parameter optimization into two simpler steps. The numerical results from Q1–Q4 allow a fairly detailed assessment of whether this decoupling actually works, and the picture that emerges is not encouraging — for reasons that go well beyond the sample size.

Stage 1 is misspecified in a way that contaminates Stage 2

The first-stage GLS fit assumes $w_i \overset{iid}{\sim} N(0, \sigma^2)$. When the true errors follow an AR(1) process with $\theta \approx 0.56$, this misspecification has two consequences that compound each other.

First, the GLS estimating equations used in Stage 1 take the form $(D^\top W D)^{-1} D^\top W y$ where D is the $n \times 3$ matrix of derivatives of μ_i and W is the weight matrix; under the iid assumption of Stage 1, $W = I$, reducing this to $(D^\top D)^{-1} D^\top y$. The efficient estimator under AR(1) errors requires replacing the identity weight matrix with $\Sigma^{-1}(\theta)$, the inverse of the AR(1) covariance matrix. Ignoring this produces a consistent but *inefficient* estimator of (α, β, γ) in large samples — though consistency itself requires regularity conditions and a unique solution to the estimating equations that cannot be verified at $n = 23$ with a highly nonlinear mean function; the Q1 covariance matrix, which assumes $W = I$, is therefore incorrect even as a large-sample approximation. The confidence intervals from Q1 are too narrow or too wide depending on the local curvature of μ_i relative to the correlation structure, and cannot be trusted for inference regardless of what happens in Stage 2.

Second, and more critically for Stage 2, the Q1 raw residuals $\hat{e}_i = y_i - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ are noisy surrogates for the true $w_i = y_i - \mu_i(\alpha, \beta, \gamma)$, subject to systematic finite-sample distortion. In large samples, as $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}) \to (\alpha, \beta, \gamma)$, the residuals do converge to the true w_i; the problem is that $n = 23$ is far too small for this asymptotic regime to be operative. The difference is

$$\hat{e}_i - w_i = \mu_i(\alpha, \beta, \gamma) - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma}),$$

which is a smooth nonlinear function of the Stage 1 parameter estimation error $(\hat{\alpha} - \alpha, \hat{\beta} - \beta, \hat{\gamma} - \gamma)$. This error does not vanish term-by-term; rather, it induces a systematic distortion in the expected autocorrelation structure of $\{\hat{e}_i\}$ relative to $\{w_i\}$, because the regression function μ_i varies smoothly with the ordered covariate. In particular, runs of positive or negative estimation error in μ_i across adjacent x_i values will inflate or deflate the lag-1 autocorrelation of $\hat{e}_i$ relative to that of w_i.

The numerical results confirm this distortion

The naive estimating equation applied to Q1 residuals returned $\tilde{\theta}_{\text{naive}} = 1.005$, which lies outside the stationary parameter space $|\theta| < 1$. This is not simply a small-sample anomaly; it is a symptom of the deterministic trend contamination described above. The sum $\sum_{i=2}^{n} \hat{e}_i$ in the numerator of $\tilde{\theta}_{\text{naive}}$ is strongly influenced by the systematic pattern visible in the Q1 residual plot (consistent positive residuals in the mid-range of x, negative at the extremes) — the same residual structure Sibbesen (1981) observed and left unmodeled, and

which motivated the AR(1) component in the first place — which is precisely the signature of the misspecified error structure from Stage 1. The naive estimating function, which has no mechanism to separate genuine autocorrelation from trend contamination, is essentially useless here.

The optimal estimating equation fares better ($\tilde{\theta}_{\text{opt}} = 0.542$) because multiplying by W_{i-1} down-weights the contribution of large residuals and focuses on local lag-1 relationships, making it more robust to the smooth global trend in $\hat{e}_i$. Nevertheless, $\tilde{\theta}_{\text{opt}}$ still deviates from the Q3 MLE of 0.560, and the direction of the difference is consistent with attenuation toward zero caused by the residual contamination.

The σ^2 estimates reinforce this: the Q1 GLS moment estimate (1.015) targets the marginal variance $\sigma^2/(1-\theta^2)$ under the iid assumption, not the innovation variance σ^2. Feeding this into a two-stage procedure would produce inconsistent starting conditions for any subsequent inference on the full model, since the two quantities are related by a factor of $1/(1-\theta^2) \approx 1.45$ at $\hat{\theta} \approx 0.56$.

Efficiency loss relative to the joint MLE

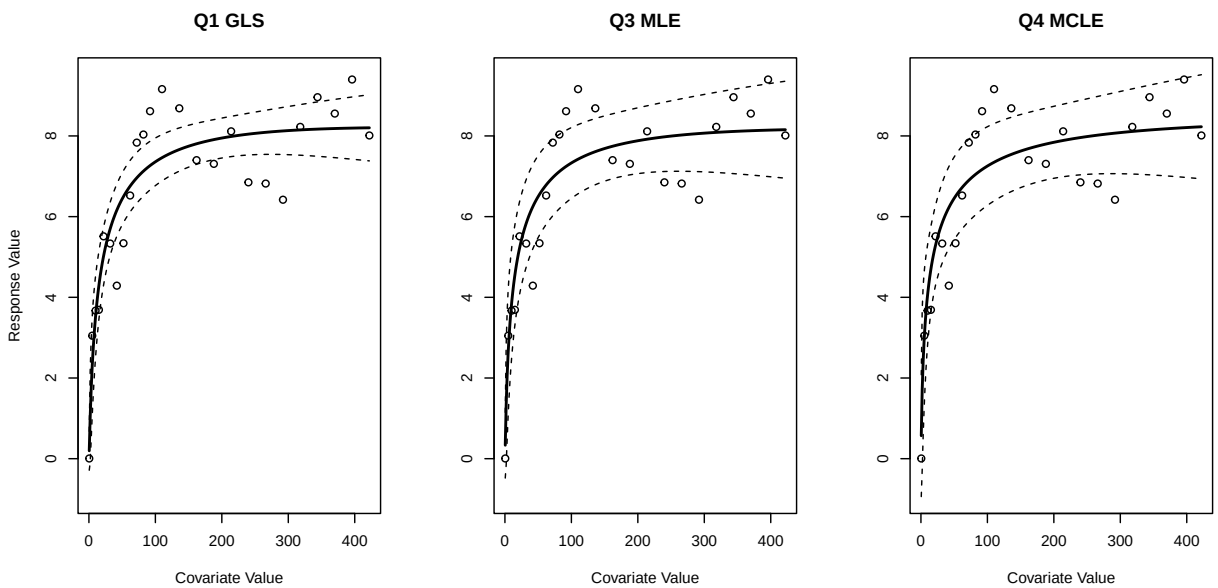

Figure 4: Fitted extended Freundlich curves with approximate 95% pointwise confidence bands: Q1 GLS (left), Q3 MLE (center), Q4 MCLE/Godambe (right).

Even if consistency of the two-stage estimator could be established under regularity conditions (it can, in principle, when $n \to \infty$ and Stage 1 gives consistent first-stage estimates), efficiency is lost relative to the joint MLE. The joint Q3 MLE accounts simultaneously for the curvature of the likelihood in all five parameters. The Stage 1 GLS, being based on a misspecified weight matrix, solves a different set of estimating equations entirely, and the resulting parameter estimates sit at a different point on the likelihood surface. The Q3 results illustrate this concretely: the MLE shifts $\hat{\alpha}$ from 0.506 to 0.739 and $\hat{\beta}$ from 1.222 to 1.028, non-trivial movements that reflect the reweighting of observations that occurs when the AR(1) structure is correctly incorporated. The profile CIs under Q3 correctly exclude inadmissible values for α and are wider for γ than under Q1 Wald, and the confidence band is wider under Q3, reflecting the increased uncertainty in (α, β, γ) once the correlated error structure is properly acknowledged. A two-stage procedure cannot reproduce this because Stage 1 is oblivious to θ.

The Q4 composite likelihood provides corroborating evidence

The Q4 pairwise conditional composite likelihood, which uses only the $n-1$ adjacent conditional pairs rather than the full joint distribution, produced estimates close to Q3 ($\hat{\theta} = 0.575$, $\hat{\sigma}^2 = 0.644$), further demonstrating that the information about θ is primarily carried by adjacent pairs and that both Q3 and Q4 are substantially self-consistent. The two-stage approach, by contrast, implicitly assumes that the information in the regression function and the information in the autocorrelation structure are separable, which they are not: the AR(1) structure of the errors means that the likelihood contributions from adjacent Y_i pairs jointly depend on all five parameters simultaneously.

Conclusion

Table 18: Small-scale estimates and mean confidence band half-widths across all methods.

Method	$\hat{\theta}$	$\hat{\sigma}^2$	Mean Band HW
Q1 GLS	—	1.015	0.623
Q2 Optimal EE	0.542	0.657	—
Q3 MLE	0.56	0.625	0.967
Q4 MCLE	0.575	0.644	1.028

Q1 GLS targets the marginal variance $\sigma^2/(1-\theta^2)$; all remaining methods target the innovation variance σ^2. Band half-widths are delta-method throughout; scalar parameter CIs for Q3 use profile intervals (Table 3).

The two-stage procedure should not be used for this model. The Stage 1 GLS fit is based on an incorrect error specification, producing residuals that are contaminated by a systematic, covariate-dependent estimation error that distorts the autocorrelation structure used in Stage 2. Consistency of the full procedure requires that this contamination vanish asymptotically, but $n = 23$ is far too small for that approximation to be credible, and the numerical evidence — particularly $\tilde{\theta}_{\text{naive}} > 1$ — suggests the distortion is materially large at this sample size. More fundamentally, the efficiency loss relative to the joint MLE is not merely a finite-sample concern: because the likelihood surface couples the regression parameters and the AR(1) parameters, decoupled estimation cannot recover the efficiency of the joint estimator even in large samples. Among the analytic estimators considered here, the joint MLE of Q3 is the most appropriate; a parametric bootstrap procedure, as suggested by the course notes for Context II applications, would be the natural next step if formal interval estimates were required.